

Word Problems — Real Life Maths

Grade 2 | Planrs Academy

Duration: 45–60 min

Topic: Addition & Subtraction

1. LESSON OVERVIEW

Learning Goal	Students will solve addition and subtraction word problems hidden inside real-life stories using a structured four-step framework.
Big Idea	Every word problem is a story with maths inside. Identifying clue words unlocks the operation needed.
Prior Knowledge	Basic addition and subtraction facts within 20; reading simple sentences.
Materials	Whiteboard / projector, printed clue-word reference cards, practice worksheets, pencils, counters (optional).

2. LEARNING OBJECTIVES

By the end of this lesson, students will be able to:

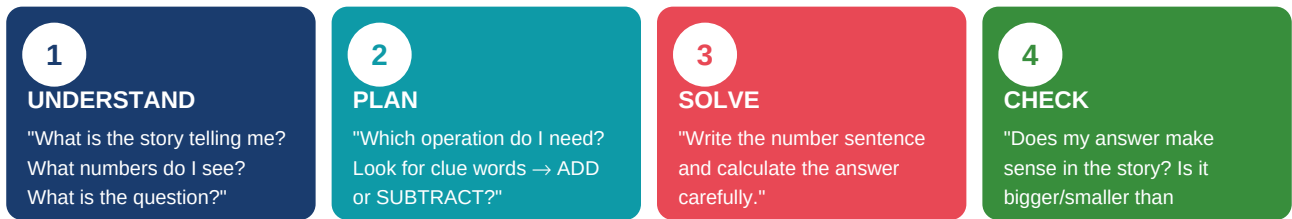
- Read a word problem and identify the key numbers and the question being asked.
- Spot addition and subtraction clue words within a story context.
- Apply the four-step framework: Understand → Plan → Solve → Check.
- Write a complete number sentence and label the answer with the correct unit.
- Verify whether an answer makes sense in the context of the story.

3. LESSON FLOW

Hook & Engagement	5 min	Present the Meera & Rohan mango problem verbally. Ask: 'What's happening in the story?'	Listen, visualise, share initial guesses. Count on fingers if needed.
Warm-Up	5 min	Flash four bare-number problems (6+7, 15–8, 9+4, 12–5) on board. Elicit quick answers.	Solve mentally or on mini whiteboards. Call out answers together.
Direct Instruction	10 min	Introduce the Four-Step Framework. Walk through the Mango Problem using all four steps on the board.	Follow along, copy framework into notebooks, predict each step.

Clue Words Lesson	5 min	Display Clue Words Reference Card. Highlight addition vs subtraction signals with real examples.	Read the card aloud together. Highlight 2 favourite clue words in their notebooks.
Clue Word Spotter	5 min	Project the Priya bangles problem and the idli problem. Guide students to spot clue words.	Underline clue words, decide operation, share thinking with a partner.
Modelled Examples	8 min	Think-aloud for Example 1 (Diwali Chaklis) and Example 2 (Arjun's Pencils) using 4 steps.	Fill in partially completed framework on guided-notes handout.
Guided Practice	7 min	Release responsibility. Display Guided Practice 1 & 2. Circulate, prompt with questions.	Work in pairs; fill in all four columns of the framework.
Independent Practice	8 min	Display Independent Problems 1 & 2. Observe; take anecdotal notes for assessment.	Solve individually and write full number sentence with label.
Consolidation & Exit	7 min	Review 'Today We Learned' summary. Administer Exit Ticket Quiz.	Circle the correct number sentence on Exit Ticket; hand in before leaving.
Home Challenge	HW	Assign home challenge: write your own word problem from real life.	Bring original problem to next lesson for class to solve.

4. THE FOUR-STEP FRAMEWORK



Teaching tip: Post a large version of this framework on the classroom wall throughout the unit. Have students point to the step they are on as they work through each problem.

5. CLUE WORDS REFERENCE CARD

ADDITION ■	SUBTRACTION ■
• altogether • total • in all • more than • plus • sum • combined • both • how many in all • added	• left • remaining • fewer • took away • spent • gave away • difference • less than • how many more • ate

6. WORKED EXAMPLES (Teacher-Led Think-Aloud)

Hook Problem — Meera and Rohan's Mangoes

Meera wants 7 mangoes. Rohan wants 5 more. How many mangoes altogether?

UNDERSTAND	PLAN	SOLVE	CHECK
Numbers: 7 and 5 Question: total mangoes	"Altogether" → ADD 7 + 5	$7 + 5 = 12$ 12 mangoes!	12 is more than 7 ✓ Makes sense!

Key teaching point: The word 'altogether' is a strong addition signal. Circle it visibly on the board.

Modelled Example 1 — Diwali Chaklis

Grandma made 13 chaklis for Diwali. She made 6 more. How many chaklis in all?

UNDERSTAND	PLAN	SOLVE	CHECK
Numbers: 13 and 6 Question: how many in all?	"In all" → ADD 13 + 6	$13 + 6 = 19$ 19 chaklis!	$19 > 13$ ✓ Makes sense!

Discuss: How is 'in all' similar to 'altogether'? Both are addition signals.

Modelled Example 2 — Arjun's Pencils

Arjun had 17 pencils. He lost 5 at school. How many pencils does he have now?

UNDERSTAND	PLAN	SOLVE	CHECK
Numbers: 17 and 5 Question: how many now?	"Lost" → SUBTRACT 17 – 5	$17 - 5 = 12$ 12 pencils!	$12 < 17$ ✓ Makes sense!

Discuss: Losing/taking away always makes the total smaller — subtraction!

7. CLUE WORD SPOTTER ACTIVITIES

Before students solve, train them to read the problem twice: first for the story, second to hunt for clue words. Use the two examples below as a shared class activity.

Priya has 9 bangles. She got 4 more. How many <u>altogether</u> ?	"more" → ADD "altogether" → ADD	ADDITION ■	$9 + 4 = 13$ bangles
There were 18 idlis. The family <u>ate</u> 7. How many are <u>left</u> ?	"ate" → SUBTRACT "left" → SUBTRACT	SUBTRACTION ■	$18 - 7 = 11$ idlis

8. GUIDED PRACTICE

Students work in pairs. Teacher circulates and uses targeted questions: 'What numbers do you see?', 'Which clue words did you spot?', 'Does your answer make sense?'

Guided Practice 1

Problem: A basket had some apples. After buying more, use the four-step framework to solve.

■ What numbers do I see? What is the question?

■ Which clue word tells me the operation?

■ Write the number sentence: ____ ■ ____ = ____

■ Check: does my answer make sense in the story?

Guided Practice 2 — Bazaar Fruits

Problem: At the bazaar, there were ____ oranges and ____ bananas. How many fruits altogether?

■ Numbers: ____ and ____

■ Clue word: _____ Operation: _____

■ ____ ■ ____ = ____

■ Does ____ make sense? Answer: ____ fruits

9. INDEPENDENT PRACTICE

Students solve individually and write a complete number sentence with a label/unit. Teacher observes and takes anecdotal notes for formative assessment.

Problem 1	Amma cooked 20 chapatis for dinner. The family ate 13 chapatis. How many chapatis were left?	Expected answer: $20 - 13 = 7$ chapatis
Problem 2	There are 11 girls and 7 boys in Kavya's class today. How many children are in the class altogether?	Expected answer: $11 + 7 = 18$ children

10. EXIT TICKET QUIZ

Riya had 16 stickers. She gave 7 to her friend. Circle the correct number sentence:

A) $16 + 7 = 23$

B) $16 - 7 = 9$ ✓ Correct Answer

C) $7 - 16 = ?$

D) $16 + 16 = 32$

Assessment guidance: Collect exit tickets as students leave. Use results to identify students who selected A (forgot the clue word 'gave'), C (reversed operands), or D (added instead) for targeted follow-up in the next session.

11. KEY TAKEAWAYS — TODAY WE LEARNED...

■ Story = Maths	A word problem is a story with maths hidden inside. Reading carefully is the first step.
■ Clue Words	Clue words tell us whether to add or subtract. Hunting for them makes the operation obvious.
■ Four Steps	Understand → Plan → Solve → Check. The framework works for every word problem.
■■ Units Always!	Always write the answer with a unit or label (mangoes, chapatis, children, pencils...).

12. HOME CHALLENGE

Can you write YOUR OWN word problem tonight using something that happened in your home? Maybe Mum cooked something, or you shared toys with a sibling, or you counted fruits in the kitchen. Write it down and bring it to class tomorrow — we will solve each other's problems together!

13. DIFFERENTIATION & TEACHER TIPS

Support	Provide counters or a number line. Give pre-filled 'Understand' and 'Plan' columns so students focus on solving and checking.
Core	Standard four-step framework. Encourage students to write two clue words before choosing an operation.

Extension	Challenge students to write their own two-step word problem, or to create a different word problem that uses the same number sentence (e.g., another story where $16 - 7 = 9$).
EAL / ESL	Pre-teach key vocabulary: 'altogether', 'remaining', 'left'. Provide bilingual clue-word cards where possible. Use drawings alongside numbers.